

Subnet Name	Subnet Address	Subnet Mask	IP Address Range
Administration	198.51.100.0	255.255.255.192	198.51.100.1 - 198.51.100.62
Sales	198.51.100.64	255.255.255.192	198.51.100.65 - 198.51.100.126
IT	198.51.100.128	255.255.255.192	198.51.100.129 - 198.51.100.190

Step 1: Find the number of addresses in range

For each IP Address Range, there are 62 addresses in the usable range. However, a subnet’s usable range doesn’t include the network and broadcast addresses. The network address is the lowest address, while the broadcast address is the highest address. Therefore, every address in between is a host address, and is within the subnet’s usable range. Based on the table below, we have a total of 64 addresses (0 - 63).

Network Address	198.51.100.0
Usable Host Range	198.51.100.1 - 198.51.100.62
Broadcast Address	198.51.100.63

Step 2: Figure out how many host bits are needed to fit 64 addresses

In binary, the number of addresses a block of host bits can hold is 2^(number of host bits). Since 2^(number of host bits) = 64, the number of host bits we can have is 6 (since 2^6 = 64).

Step 3: Calculate the Subnet Mask

Since the address is IPv4, it has a total of 32 bits (four octets). This means 26 of those bits are the network portion and 6 of them are the host portion. So our subnet mask is /26 because our first 26 bits identify the network and the last 6 bits identify the hosts within the subnet.

In binary, our subnet mask is 11111111 . 11111111 . 11111111 . 11000000.
Converted back to decimal, the subnet mask is 255.255.255.192 for each of our three subnets.

Step 4: Calculate the Subnet Address

To calculate the Subnet Address for each subnet, we need to use the AND operation on each network address and the subnet mask, with both in binary. The AND operation means that each bit of the subnet address is 1, iff (if, and only if), the corresponding bits for the network address and subnet mask are both 1. The table below shows the operation on the Administration subnet.

Administration Subnet Mask	11111111 . 11111111 . 11111111 . 11000000
Administration Network Address	11000110 . 00110011 . 01100100 . 00000001
Administration Subnet Address	11000110 . 00110011 . 01100100 . 00000000

Therefore, the binary address in decimal is 198.51.100.0 for the Administration subnet. The binary addresses for the other subnets are calculated below. They are also added to the full table at the top of the page.

Sales Subnet Mask	11111111 . 11111111 . 11111111 . 11000000
Sales Network Address	11000110 . 00110011 . 01100100 . 01000001
Sales Subnet Address	11000110 . 00110011 . 01100100 . 01000000

Sales Subnet Address (Decimal): 198.51.100.64

IT Subnet Mask	11111111 . 11111111 . 11111111 . 11000000
IT Network Address	11000110 . 00110011 . 01100100 . 10000001
IT Subnet Address	11000110 . 00110011 . 01100100 . 10000000

IT Subnet Address (Decimal): 198.51.100.128